

COOK, MN, USA

WMO: 726549

Lat: 47.822N

Lon: 92.689W

Elev: 405

StdP: 96.56

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54905

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-28.0	-35.9	0.1	-32.4	-32.2	0.2	-28.6	8.1	-10.8	7.4	-10.6	0.5	270	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	29.0	20.7	27.6	19.7	26.3	18.7	22.7	27.2	21.4	25.9	20.2	24.5	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.7	25.6	19.8	15.2	24.2	18.7	14.2	23.0	68.9	27.2	63.8	26.1	60.1	24.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	5.6	DB	-35.5	32.1	2.8	1.4	-37.5	33.1	-39.2	33.9	-40.7	34.7	-42.7	35.8	(4)
(5)			WB	-35.5	24.3	2.7	1.0	-37.5	25.0	-39.1	25.6	-40.7	26.1	-42.7	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-13.2	-11.2	-3.9	3.8	10.8	15.7	18.7	17.3	13.1	5.7	-2.2	-10.2		
	DBStd	12.59	8.06	7.90	7.52	5.36	4.86	3.77	3.30	3.41	4.83	4.88	6.23	7.39		
	HDD10.0	3158	720	593	432	195	47	3	0	1	23	152	367	625		
	HDD18.3	5417	978	827	689	436	238	96	35	61	168	392	616	883		
	CDD10.0	893	0	0	2	9	73	174	271	227	116	19	1	0		
	CDD18.3	111	0	0	0	0	6	17	48	29	11	1	0	0		
	CDH23.3	1244	0	0	0	4	93	200	526	313	101	7	0	0		
	CDH26.7	268	0	0	0	1	18	37	127	65	19	1	0	0		
(14) Wind	WSAvg	2.1	2.2	2.1	2.2	2.3	2.4	2.0	1.8	1.8	2.1	2.3	2.3	2.0		
(15) Precipitation	PrecAvg	697	19	17	29	53	80	108	98	83	86	65	35	24		
	PrecMax	914	46	48	78	138	166	194	281	138	205	133	92	40		
	PrecMin	512	4	2	10	16	28	27	41	31	18	21	3	6		
	PrecStd	88	10	11	17	29	33	37	52	33	42	29	22	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	8.0	17.6	23.0	28.8	30.1	32.2	31.1	28.9	23.8	16.1	6.0		
		MCWB	2.1	4.1	11.4	13.1	18.3	21.1	22.6	22.6	19.9	15.3	10.9	3.1		
	2%	DB	2.2	5.9	12.7	19.1	26.2	27.5	29.1	28.0	26.3	20.0	12.4	2.9		
		MCWB	0.7	2.8	7.6	10.4	16.3	19.1	21.1	20.5	18.9	13.6	9.4	1.4		
	5%	DB	0.0	2.7	9.0	17.3	22.9	26.0	27.7	27.1	23.8	17.2	8.9	1.3		
		MCWB	-0.7	0.9	5.5	9.0	14.2	17.9	20.2	19.8	18.0	12.2	6.0	0.4		
	10%	DB	-2.2	1.0	6.3	14.0	21.0	23.7	26.4	24.9	22.1	13.8	7.0	-0.1		
		MCWB	-2.9	-0.3	3.0	7.3	13.4	16.9	19.1	18.4	17.0	9.9	4.8	-0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	5.4	13.9	14.5	20.5	23.0	24.4	23.9	22.2	17.2	12.4	3.4		
		MCDB	4.4	7.6	16.2	21.3	25.7	27.8	28.8	27.7	26.7	20.5	14.0	4.8		
	2%	WB	0.9	3.0	8.7	11.8	18.0	20.9	23.1	22.2	20.4	14.5	8.9	1.9		
		MCDB	1.8	5.1	11.6	16.8	22.9	25.2	27.2	26.9	24.0	19.1	12.2	3.1		
	5%	WB	-1.0	1.2	5.8	10.0	16.1	19.5	21.6	20.7	19.1	12.6	6.7	0.6		
		MCDB	-0.2	2.8	9.2	16.1	21.3	23.7	26.1	25.2	22.5	16.2	8.5	1.4		
	10%	WB	-2.7	-0.8	3.5	7.9	14.3	18.0	20.3	19.6	17.6	10.8	4.4	-0.9		
		MCDB	-2.3	0.7	6.1	13.3	18.7	21.9	25.0	23.5	20.7	13.3	6.6	-0.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.7	12.5	12.6	13.3	12.6	12.9	12.8	12.2	9.9	8.4	9.3		
		MCDBR	10.3	12.0	13.8	17.9	17.7	14.8	14.4	14.7	14.8	14.8	12.3	9.8		
		MCWBR	8.9	9.1	9.2	10.4	9.2	7.8	7.3	7.7	8.4	9.2	8.7	8.3		
		MCDBR	9.6	10.1	12.7	15.8	13.8	12.1	12.7	12.3	12.0	12.9	11.1	9.0		
	5% WB	MCWBR	8.4	7.8	9.0	9.9	8.0	7.4	7.5	7.2	7.6	8.6	8.4	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.264	0.314	0.345	0.359	0.372	0.390	0.373	0.356	0.307	0.282	0.260		
	taud		2.359	2.326	2.340	2.382	2.402	2.383	2.355	2.431	2.454	2.577	2.523	2.395		
	Ebn at Noon		845	909	905	908	904	890	868	869	852	855	808	785		
	Edh at Noon		68	88	104	111	113	116	118	105	93	70	59	59		
(44) All-Sky Solar Radiation	RadAvg		1.38	2.53	3.72	4.68	5.25	5.59	5.97	5.06	3.75	2.21	1.26	0.97		
	RadStd		0.17	0.25	0.35	0.46	0.37	0.43	0.29	0.37	0.30	0.30	0.16	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page